

AI Assignment: You are an experienced Web Development instructor.

Instructions: Answer all questions. Submit as PDF.

Questions:

1. Explain the role of HTML in web development and describe how it interacts with CSS and JavaScript to create dynamic web pages.
2. Create an example webpage using HTML, CSS, and JavaScript that includes a form with validation. Describe how each technology contributes to the functionality of the page.
3. Compare and contrast client-side scripting (JavaScript) with server-side programming languages like PHP or Node.js in terms of their roles and capabilities within web development.
4. Describe the Document Object Model (DOM) and explain its importance in JavaScript for manipulating webpage elements dynamically. Provide an example of a DOM manipulation task using JavaScript code.
5. Explain how HTTP requests work between a client browser and a server, including the process from request to response. Discuss the role of headers, status codes, and common HTTP methods.

Due Date: 12/17/2025
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